

AESH SUSTAINABILITY HACKATHON 2026

CHALLENGE 3 FINALIST GUIDE

Sustainable Sonar Systems for Marine & Climate Protection

FINAL PHASE REQUIREMENTS AND DEADLINES

Onsite Final - Egypt | 6-7 July 2026

COMMITTEE REVIEW DRAFT

Final Phase objective

Demonstrate useful marine sensing while reducing acoustic disturbance, unnecessary active transmission, or energy use.

1. What Your Team Must Build

- One marine/climate protection use case and a clear reason acoustic sensing is appropriate.
- Active, passive, or hybrid sonar mode plus realistic deployment, range, noise, depth, and propagation assumptions.
- An implemented signal-processing chain from input/acquisition to detection, classification, or estimation.
- A live executable pipeline or simulation producing interpretable sensing output.
- A fair conventional/aggressive baseline compared with the eco-friendly operating configuration.
- Quantified acoustic or active-use reduction plus sensing-performance trade-off.
- Validation across multiple noise, range, source-level, duty-cycle, or threshold cases.

2. Required Final Deliverables

Deliverable	Requirement
Updated Project Repository	Accessible to judges; organized files; README with setup, execution, parameters, assumptions, and reproduction steps.
Working Proof of Concept	Hardware, software, executable simulation, or validated analytical model demonstrating the core function.
Final Evidence Sheet	One page: problem, core function, baseline, improved case, test conditions, primary technical KPI, primary sustainability KPI, limitation, and repository link.
Final Presentation	No fixed slide count. Use the number of slides that fits a 5-minute presentation and keeps text readable.
Live Demo	Maximum 5 minutes; must show the core function and main evidence.
Backup Demo Video	60-90 seconds; used only if a genuine technical failure occurs.
Equipment and Safety Declaration	All power, hardware, RF, acoustic, mechanical, battery, and special-space needs must be declared and approved.
AI / External Resource Disclosure	List meaningful AI assistance, libraries, datasets, prior work, datasheets, standards, and external technical sources.

3. Challenge-Specific Repository Evidence

- Sonar code/notebook and parameter files.
- Data or signal-generation description.
- Propagation/noise and processing-chain explanation.
- Baseline and eco-friendly operating comparison.
- Labeled sensing, acoustic-impact, energy, and trade-off plots.

4. Recommended Primary KPIs

- Source-level or exposure proxy reduction
- Duty cycle / active-transmission reduction
- Energy or unnecessary-use reduction
- Detection range/accuracy/coverage retained

5. Official Deadlines

Time zone

All deadlines are in Egypt Summer Time (EEST, UTC+3). A written update from the organizing committee overrides this table only if issued officially.

Deadline	Required action
17 June 2026 - 23:59 EEST	Finalist confirmation, onsite attendance confirmation, and final team roster.
20 June 2026 - 23:59 EEST	Repository-access confirmation plus Equipment and Safety Declaration.
23 June 2026 - 23:59 EEST	One-page Final Phase Implementation Plan: scope, core function, KPI, test plan, and main risk.
30 June 2026 - 23:59 EEST	Progress proof: 60-90 second progress video, one technical result, and one sustainability result.
3 July 2026 - 23:59 EEST	Final digital package: updated repository, evidence sheet, presentation, backup demo, and disclosures.
6 July 2026	Onsite Day 1: opening, workshops, preparation, networking, and related activities. Detailed agenda to be announced later. No official judging.
7 July 2026 - 12:00 EEST	Final repository tag/file freeze and final presentation confirmation.
7 July 2026 - 13:30 EEST	Official Final Phase judging begins.

6. Final-Day Format

- Day 1 (6 July): opening, workshops, preparation, networking, and related activities. Detailed agenda will be announced later. No official judging.
- Day 2 judging begins at 13:30.
- Each team receives 15 minutes: 5-minute presentation, 5-minute live demo, 4-minute Q&A, and 1-minute scoring/transition.
- The panel includes the designated Technical Judge for your challenge and two Sustainability Reviewers who evaluate every challenge.
- Final ranking uses 75% technical and 25% average sustainability. Only First, Second, and Third Place are awarded.

7. What Is Not Sufficient

- Repeating the Phase 1 concept without new implementation or validation.
- A promotional video with no technical evidence.
- An unexplained percentage improvement or a baseline that performs a different task.
- Screenshots without runnable files, parameter values, units, or interpretation.
- A complex project that cannot demonstrate one stable core function.
- Using AI-generated or external work that the team cannot explain, verify, or disclose.

8. Rules and Safety

- Continue the same challenge and core project qualified in Phase 1. Scope may be refined, but the project may not be replaced.
- Only registered team members may present, operate the demo, and answer questions.
- Unauthorized RF transmission, high-power acoustic transmission, hazardous batteries, lasers, chemicals, pressure systems, or unsafe mechanical motion are prohibited.
- Teams retain responsibility for originality, licensing, data accuracy, and full understanding of their submission.
- The organizing committee may stop unsafe or non-compliant demonstrations.